

Read each of the questions carefully and answer showing all steps of your solution

Final Answers with no clear solution get ZERO

Q1- Find the root of the function: $f(x) = \sin(x) + x^2 + x - 10$ using: (12 marks)

(a) The *Bisection Method*. Perform two iterations to find the root, use the interval (2, 4)

$$x_{r3} = \dots\dots\dots$$

(b) The *Newton Raphson Method*. Perform two iterations to find the root, assume $x_0 = 2$.

$$x_2 = \dots\dots\dots$$

Note: work to 5 significant figures

Solve the following systems of linear equations (12 marks)

(a) Use the *Cramer's Rule* to solve the following two simultaneous equations, determine x_1 and x_2 :

$$x_1 + x_2 = 25$$

$$6x_1 - 3x_2 = 6$$

$$x_1 = \dots\dots\dots, x_2 = \dots\dots\dots$$

(b) Use the *Gauss Elimination Method* to solve the following three simultaneous equations.
Determine the values of x, y, z :

$$-3x - y + 2z = 25$$

$$4x + 2y + z = 16$$

$$5x + 3y - \frac{z}{2} = 3$$

$$x = \dots\dots\dots, y = \dots\dots\dots, z = \dots\dots\dots$$

Q3- Use the *Bairstow Method* applying first iteration, to find the roots of the polynomial: (6 marks)
 $f(x) = x^3 - 2x^2 + x - 2$. Use $(x^2 - 0.5x + 1)$, as a starting factor.

Given the results of the first synthetic division are:

$$b_3 = 1, b_2 = -1.5, b_1 = -0.75, b_0 = -0.875$$

End of Questions

⇒ (Q.2)

(Q.) ⇒ sol: $f(x) = \sin(x) + x^2 + x - 10$ / $x_l = 2, x_w = 4$

Zero ~~iteration~~ iteration

$$\Rightarrow x_r = \frac{2+4}{2} = 3 \quad / \quad f(x_l) = -3.0907 \quad / \quad f(x_w) = 9.2432$$

$$f(x_r) = 2.1411$$

20x1
Rad

⇒ new interval → $x_w = 3$ / $x_l = 2$

~~1st iteration~~ {2st iteration}

$$x_{r_1} = \frac{3+2}{2} = 2.5 \quad / \quad f(x_l) = -3.0907 \quad / \quad f(x_w) = 2.1411$$
$$f(x_r) = -0.6515$$

2nd iteration
⇒ new interval → $x_l = 2.5$ / $x_w = 3$

$$\Rightarrow x_{r_2} = \frac{2.5+3}{2} = 2.75$$

Amazing

⑥ $\Rightarrow$ sol:

$$f(x) = \sin(x) + x^2 + x - 10 \quad / \quad x_0 = 2$$

$$f'(x) = \cos(x) + 2x - 1$$

in π
rad

$$x_{i+1} = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} \Rightarrow x_1 = 2 - \frac{-3.0907}{2.5838}$$

$$x_1 = 3.1962$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} \Rightarrow x_2 = 3.1962 - \frac{3.3573}{4.3939}$$

$$x_2 = 2.4321$$

~~✗~~

{Q2.2}

$$\textcircled{a} \quad \begin{aligned} x_1 + x_2 &= 25 \\ 6x_1 - 3x_2 &= 6 \end{aligned}$$

$$\Rightarrow \text{soln} \quad [A] \begin{bmatrix} 1 & 1 \\ 6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 25 \\ 6 \end{bmatrix}$$

~~soln~~ $\Rightarrow$ To find $[X]$:- $\det[A] = (1 \times -3) - (1 \times 6)$
 $= -3 - 6 = -9$

$$[A_x] = \begin{bmatrix} 25 & 1 \\ 6 & -3 \end{bmatrix} \quad \Rightarrow X = \frac{\det[A_x]}{\det[A]}$$

$$\det(A_x) = (25 \times -3) - (1 \times 6)$$
$$= -81$$

$$X = \frac{-81}{-9} = 9 \Rightarrow \boxed{X = 9}$$

$\Rightarrow$ To find (y) :-

$$[A_y] = \begin{bmatrix} 1 & 25 \\ 6 & 6 \end{bmatrix} \quad \rightarrow \det(A_y) = (1 \times 6) - (25 \times 6)$$
$$= -144$$

$$\Rightarrow y = \frac{\det(A_y)}{\det[A]} = \frac{-144}{-9} = 16$$

$$\boxed{y = 16}$$

⇒ (b) solⁿ

$$-3x - y + 2z = 25$$

$$4x + 2y + z = 16$$

$$5x + 3y - \frac{z}{2} = 3$$

$$\Rightarrow \left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 4 & 2 & 1 & 16 \\ 5 & 3 & -0.5 & 3 \end{array} \right] \quad 4R_1 + 3R_2$$

$$\left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 0 & 2 & 11 & 148 \\ 5 & 3 & -0.5 & 3 \end{array} \right] \quad 5R_1 + 3R_3$$

$$\rightarrow \left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 0 & 2 & 11 & 148 \\ 0 & 4 & 8.5 & 1134 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} -3 & -1 & 2 & 25 \\ 0 & 2 & 11 & 148 \\ 0 & 0 & -13.5 & -162 \end{array} \right]$$

↗ -2R₂ + R₃

$$0(x) + 0(y) - 13.5z = -162$$

$$\Rightarrow \boxed{z = 12}$$

$$\Rightarrow 0(x) + 2(y) + (11 \times 12) = 148$$

$$\Rightarrow \boxed{y = 8}$$

$$\Rightarrow -3(x) - 8 - (2 \times 12) = 25 \Rightarrow \boxed{x = -19}$$

→ starting factor.

← حساب هو ملاحق من اول قسمات لانا ملاحق ملاحق ← (b)

2)

	1	-1.5	-0.75	-0.875
P_1	0.5	0.5	-0.5	-1
P_2	-1	-1	-2.25	
C_3				
C_2				
C_1				

بسم
بالقسط
~~بسم~~
بسم
بسم

→ بتفرد
فاطمية
التي ما بقا 2
بالعارة → Co

$$\Rightarrow c_2 S_p + c_3 S_q = b_1 \quad \dots \textcircled{1}$$
$$\Rightarrow (a_1 - b_1) S_p + c_2 S_q = b_0 \quad \dots \textcircled{2}$$

$$\Rightarrow -s_p + s_q = -0.75 \quad \text{--- 1}$$

$$\Rightarrow (-2.25 + 0.75) s_p - s_q = -0.875 \quad \text{--- 2}$$

$\left. \begin{array}{l} \text{Lecture 1} \\ s_p(x) \leftarrow \\ \text{Lecture 2} \\ s_q(y) \leftarrow \end{array} \right\} \text{نوعان من المتغيرات العشوائية}$

$$\Rightarrow s_p = 0.65 / s_q = -0.1$$

المطلوب

$$\Rightarrow x^2 + (p + s_p)x + (q + s_q)$$

$$\Rightarrow x^2 + (-0.5 + 0.65)x + (\pm 1 + (-0.1))$$

$$\Rightarrow x^2 + 0.15x + 0.9 = 0$$

$$\Rightarrow x = \frac{-0.075 \pm 0.9457}{1}$$

$\downarrow$ (A) $\downarrow$ (B)